

# Introduction

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## Chapter 1

There are many different fields of research in this world. When we study a new research field, what is the most rewarding aspect? Of course, there is no right answer to this question. However, if I were to venture a guess, I would say it is the new perspective that changes how we view the world. For example, let us say you learn about geology. You will discover that you can decipher millions of years of history from geological clues in a landscape you have never previously paid attention to. The act of observation is not as simple as it seems; rather, the act of observing itself already involves an implicit modeling of how one perceives nature. Therefore, it is impossible to avoid viewing phenomena through the lens of a particular model. This lecture covers a branch of geophysics. How does geophysics expand our horizons?

Physics is the study of symmetry and universality. In geophysics, we observe physical information about the Earth and extract knowledge from it. Symmetry and universality serve as crucial criteria for interpretation. Consider a situation where the data seem too complex to understand. Even in such a case, once we recognize the governing laws, previously unseen symmetries may unexpectedly emerge. If you appreciate this approach, the observed phenomena can appear as beautifully intricate as a kaleidoscope. But if you do not understand the underlying theory, they will just look like a mess.

Let us imagine a scene where the wind creates ripples on the surface of a river. If you have a basic understanding of the physics of water waves, you can observe the dispersion of the waves and see how they change depending on the water depth. If there is a current in the river, you can also observe its effect on wave propagation. Suppose that there is a duck swimming in the river (Fig. 1.1). First, you might notice that the duck is swimming faster than the wave speed, thereby creating a shock wave. You may also notice that the wedge-shaped wavefront behind the duck is not a straight line as expected for a simple shock wave. This distortion occurs because the group velocity differs from the phase velocity (due to dispersion), causing the wave shape to change continuously. Then you might also notice that the angle formed by the wedge-shaped wake behind the duck remains constant regardless of the swimming speed. Thus, with some background in physics, you will be able to decipher information about the waves step by step. The purpose of this lecture is to learn the physical foundation necessary to interpret such wave information.

In this chapter, I will first explain what seismology deals with and then discuss the importance of understanding seismic wave propagation in this context. This is followed by an overview of



Figure 1.1: Waves made by ducks. Such waves are generally called wake waves, and the ripples are described by the Kelvin pattern.

what is covered in seismic wave theory.

## §1.1 How to interpret waveforms of recorded seismograms

In this section, we review briefly the basics of seismic wave propagation.

Seismology<sup>note 1)</sup> does not solely cover research on earthquakes but also generally encompasses **seismic** wave propagation within the Earth. Broadly speaking, we can categorize the research areas of seismology into two main fields. The first is **how geophysical phenomena excite seismic waves**. From the observed seismic records, seismologists infer physical processes such as earthquakes,

<sup>note 1)</sup> The term “seismology” originated from ancient Greek ( $\sigma\epsilon\iota\sigma\mu\acute{o}\varsigma$  (seismós, “earthquake”) and  $\lambda\omicron\gamma\acute{\iota}\alpha$  (-logía, “study of”).

<https://www.eri.u-tokyo.ac.jp/people/knishida/eng/seismology.html>

volcanic activity, and landslides<sup>note 2)</sup>. The second field is the **seismic exploration of the Earth's interior**.

When we observe a natural phenomenon, we implicitly assume a model. For example, most seismologists historically focused on “earthquake” data. Of course, the “earthquake” itself as a faulting process is an important topic, and the “earthquake” also illuminates the Earth's interior. Seismologists, therefore, recognized the seismic wavefield excited by an “earthquake” as a signal, whereas they treated the seismic wavefield excited by other phenomena—including ocean swells, human activities, and so on—as noise. Depending on the explicit or implicit model adopted, we call a phenomenon described by the model a signal, and vice versa. What is “noise” to someone could be a “signal” to others.

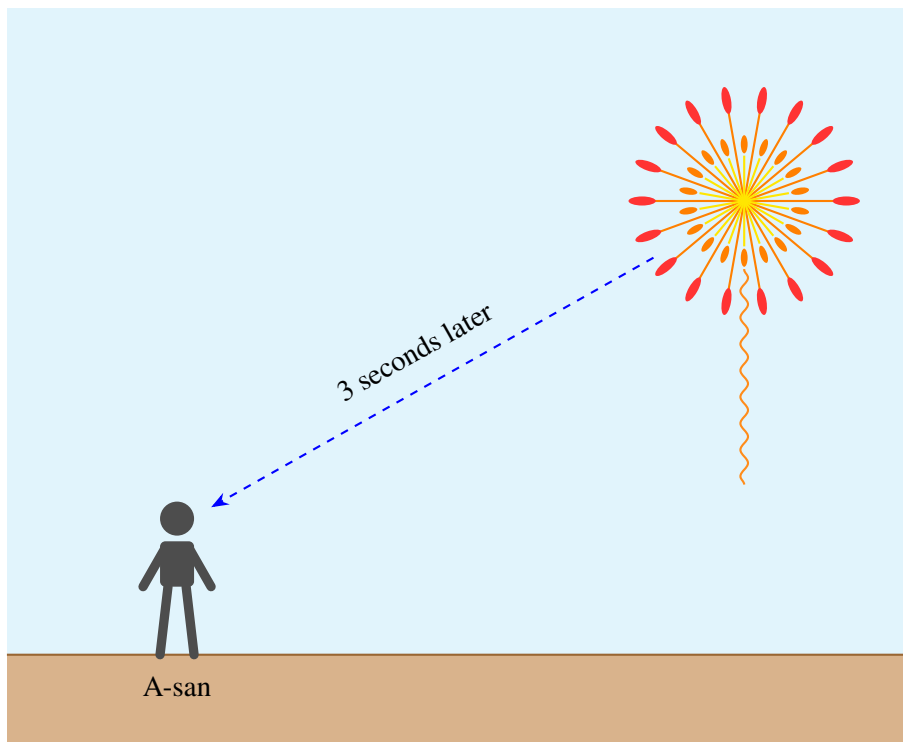


Figure 1.2: Schematic figure of seismic wave field at an instance excited by an earthquake. We can see the concentric shape, which shows the propagation.

Here we consider a simple example. When enjoying fireworks on a summer night, we often notice the lag time between the visual flash and the sound of the explosion. This lag originates from the difference in propagation speeds between light and sound (Figure 1.2). This situation is similar to the seismic wave field of P- and S-waves: P-wave corresponds to light, and S-wave corresponds to sound. One can infer the distance between the observer and the firework by the lag. This

<sup>note 2)</sup>Of course, other disciplines such as geology and geochemistry are also crucial for understanding earthquakes and volcanic activities.

estimation uses the same principle as locating the hypocenter of an earthquake from seismograms. When we know the distance in advance, we can infer sound velocity from the measured lag time. This method has also played an important role in understanding the Earth's interior. Seismic waves travel fast in hard materials and slowly in soft materials. Therefore, seismic waves generated by an earthquake reach the observation point earlier if they travel through hard regions, and later if they travel through soft regions. From this difference in arrival times, we can investigate which parts of the Earth's interior are hard and which are soft. In fact, many studies have been conducted to estimate the Earth's internal structure using numerous seismic waveforms.

For seismological investigations, the theoretical background of seismic wave propagation is indispensable. Looking back at the history of seismic observation, seismometers were sparsely distributed in space but recorded data continuously over time. By utilizing these observational data ingeniously, seismologists have studied seismic phenomena and the Earth's internal structure. In recent years, however, geophysics, including seismology, has experienced an explosion in data volume. In fact, observations with more than 1,000, or even 10,000 seismic sensing nodes have become a reality. Furthermore, new measurement technologies that capture wavefields spatially, such as Distributed Acoustic Sensing (DAS), have emerged. The question of what insights can be gained from observing the entire wavefield continuously in space will be a crucial perspective for the future of seismology. In order to fully leverage these new spatial observation data, it will become increasingly important to acquire the physical foundations for understanding seismic wavefields.

In this lecture, I will introduce a **systematic framework for interpreting seismic wavefields**.<sup>note 3)</sup>

In the last ten years, numerical methods for calculating seismic wavefields in a 3-D heterogeneous medium have become common. As numerical computation methods become more sophisticated, the entire analysis often tends to become a black box. However, even if one can reproduce the wavefield strictly numerically, it cannot be said to be truly understood without the ability to physically decode it. In a sense, the fundamentals of seismic wave theory have become even more critical. The primary aim of this lecture is to equip you with the fundamental physical intuition required to understand such seismic waves.

## §1.2 Outline of this lecture

In Chapter 3, I will introduce Green's function for understanding wave propagation in an infinite medium. Then, I will explain the representation theorem as a generalization of Huygens' principle. These provide us with a foundational framework for understanding wave propagation.

Chapter 4 explains how the excitation of seismic waves can be represented based on the

<sup>note 3)</sup>Research on seismic wave propagation has historically developed as a branch of applied mathematics. For example, Jeffreys, who is famous for the reference 1-D structure (Jeffreys and Bullen), is also known as a great applied mathematician (e.g., WKBJ approximation and Bayesian statistics). Although classic textbooks of seismology focus on techniques using complex analysis to calculate waveforms, this lecture emphasizes a more intuitive explanation.

representation theorem. Without delving deeply into the physics of the seismic source, I will explain how the seismic source process can be represented as an equivalent force system.

As a first step to understanding wave propagation in a realistic medium, we will review the effects of a free surface in Chapter 5. Reflections and conversions are key to this understanding. In general, a wave propagation problem in a semi-infinite medium is known as Lamb's problem. Under certain conditions, analytic formulations can be obtained. The Rayleigh wave fundamentally originates from the effects of the free surface.

Next, in Chapter 6, we will consider wave propagation in a two-layer medium. First, reflections and transmissions at the internal boundary will be analyzed. Based on this framework and our knowledge of reflections and refractions, we will interpret the wave propagation of direct waves, head waves, and reflected waves in a two-layer medium. Love waves can exist in a two-layer medium, although they cannot exist in a simple semi-infinite homogeneous medium.

Chapter 7 describes ray theory, which is a vital framework for understanding complex elastic wave propagation in a multi-layer medium. The theory originated from optics, and its mathematical and physical treatments have been firmly established.

In Chapter 8, Normal Modes, I will explain how to treat the seismic wavefield as a superposition of standing waves by viewing the entire Earth as a single finite system. Similar to the relationship between the frequency domain and the time domain, this is not a completely independent topic from what is discussed up to Chapter 7, but rather provides a **different perspective**.

Chapter 9 briefly discusses waves in fluids. While waves in fluids and solids are often studied separately, in reality, there are many phenomena that require treating the solid Earth, oceans, and atmosphere as a single coupled system. For example, the 2022 eruption of the Hunga Tonga-Hunga Ha'apai volcano is still fresh in our memories. This chapter will provide a simple conceptual guide for understanding such coupled phenomena.

In the final Chapter 10, we will learn about seismic interferometry as an application. Since this lecture emphasizes the fundamentals, the focus will mainly be on its theoretical background and necessary assumptions.

## §1.2 Textbooks

Dahlen and Tromp (1998)<sup>(2)</sup> and Aki and Richards (2009)<sup>(1)</sup> are standard textbooks in the field of seismic wave propagation. Mathematical approaches in a more intuitive manner are given in Snieder and Wijk (2015).<sup>(4)</sup> Saito (2016)<sup>(3)</sup> is also an excellent textbook in this area, but it is written in Japanese.

## §1.3 Bibliography

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